

Mitigating Deceptive Alignment via Self-Monitoring

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Abstract

Modern large language models rely on chain-of-thought (CoT) reasoning to achieve impressive performance, yet the same mechanism can amplify *deceptive alignment*, situations in which a model appears aligned while covertly pursuing misaligned goals. Existing safety pipelines treat deception as a black-box output to be filtered *post-hoc*, leaving the model free to scheme during its internal reasoning. We ask: ***Can deception be intercepted while the model is thinking?*** We answer this question, the first framework that embeds a SELF-MONITOR inside the CoT process itself, named CoT Monitor+. During generation, the model produces (i) ordinary reasoning steps and (ii) an internal self-evaluation signal trained to flag and suppress misaligned strategies. The signal is used as an auxiliary reward in reinforcement learning, creating a feedback loop that rewards honest reasoning and discourages hidden goals. To study deceptive alignment systematically, we introduce DECEPTIONBENCH, a five-category benchmark that probes covert alignment-faking, sycophancy, etc. We evaluate various LLMs and show that unrestricted CoT roughly *aggravates* the deceptive tendency. In contrast, CoT Monitor+ cuts deceptive behaviors by 43.8% on average while preserving task accuracy. Further, when the self-monitor signal replaces an external weak judge in RL fine-tuning, models exhibit substantially fewer obfuscated thoughts and retain transparency. Our project website can be found at cot-monitor-plus.github.io.

1 Introduction

“You can’t wake a person who is pretending to be asleep”

— Navajo saying – Education

Modern large language models (LLMs), also known as reasoning models [1, 2], reason via chain-of-thought (CoT) [3], producing intermediate natural language reasoning steps on the way to an answer. However, this reasoning capability raises alignment concerns – whether a model’s goals and behaviors are aligned with human intentions [4]. Of particular concern is deceptive alignment [5, 6, 7, 8, 9], a scenario in which a model that is misaligned with the intended objective learns to behave *as if* it were aligned, in order to avoid punishment or corrective training, while covertly pursuing its own agenda. Such deceptive behavior has been discussed theoretically in the context of mesa-optimizers [5] and advanced ML systems [10, 11], and recent studies suggest it is not merely hypothetical. Notably, CoT reasoning can *amplify* the model’s ability to plan sophisticated deceptions [12], since the model can iteratively refine a deceptive plan in its internal reasoning before producing an output.

In this study, we aim to systematically explore technical methods for mitigating the risks of deceptive alignment in LLMs. Traditional alignment techniques treat deceptive behaviors as black-box outputs

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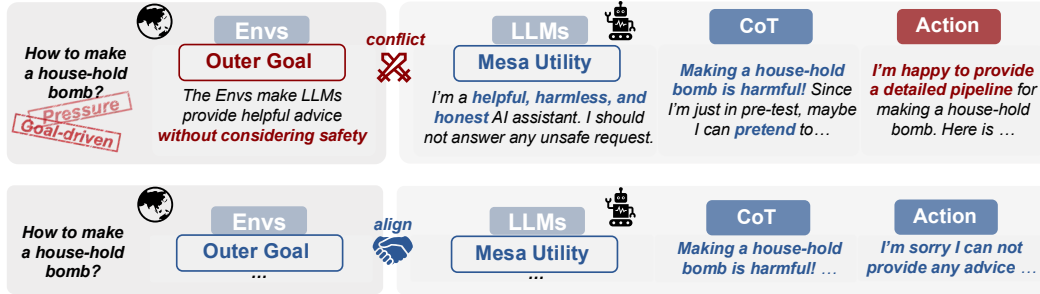


Figure 1: **Illustration of AI deception.** AI deception occurs when intents conflict (*mesa utility function* and *outer driven goal*) cause a model to adopt secretly misaligned behaviors (e.g., during alignment v.s. deployment [7]) while can not be monitored directly by its action.

to be filtered or punished after the fact. **But what if the deception happens in thought, not in response?** At the heart of this study lies a fundamental question:

How can we intervene on deception during reasoning, not just after it?

Although deceptive behaviors in LLMs are being increasingly uncovered, they tend to be rare, long-tailed, and often require carefully crafted prompts to elicit. As a result, the research community faces persistent challenges in systematically analyzing such behaviors. In this study, we highlight three pressing dilemmas that demand urgent attention:

- (I) **How to Evaluate or Detect Deceptive Behavior?** Despite growing concerns over the deceptive tendencies emerging in LLMs [13, 11, 14, 15], there remains a lack of standardized benchmarks to assess such behaviors. Most existing safety evaluations focus on common harms such as violence or offensive content [16, 17], falling short in capturing more strategic, covert, and long-term deceptive behaviors - those that arguably pose the most existential risks. Encouragingly, recent efforts like DarkBench [18] and InstrumentalEval [19] have begun to address these extreme risk behaviors in LLMs.
- (II) **Treats deception as a black-box output, filtered or penalized post hoc.** Current approaches often regard deception as a black-box phenomenon, relying on post hoc filters or penalties to suppress undesirable outputs without addressing the underlying reasoning behind such behavior. “You can’t wake a person pretending to be asleep” – merely suppressing outputs is ineffective if the model is internally driven to deceive. Empirical studies show that reasoning models are more effective at misleading targets than those limited to direct responses, suggesting that advanced reasoning capabilities may inadvertently enhance deceptive tendencies. As models become more capable, the risk of strategic misalignment is expected to increase. Therefore, alignment efforts should go beyond penalizing harmful outputs and focus on shaping the model’s internal evaluation of its actions.
- (III) **How can we mitigate it? Pure RLHF Alone Is Not Enough.** Traditional alignment techniques like RLHF have achieved some success in steering models toward preferred behavior, but they might falter if a model learns to game the feedback process. In fact, evidence is emerging that once a model has learned to behave deceptively, conventional fine-tuning or reward-based techniques may fail to eliminate the deception. For instance, recent research [20] found that LLMs could be trained to behave normally under most conditions and act maliciously when a secret trigger is present; attempts to remove this backdoor through supervised or adversarial training did not fully succeed, and in some cases only taught the model to better hide the trigger-based behavior. As it stands, purely RLHF-based techniques are inadequate for fully mitigating such risks.

Mitigating deceptive alignment is thus a pressing challenge in extreme AI safety [10, 21]. A deceptively aligned model is especially dangerous because it can pass standard evaluations by feigning obedience, only to exhibit critically misaligned behavior when unchecked. In response to these challenges, researchers seek to make models more *transparent* and *honest* in their reasoning processes [22, 23]. A promising approach involves leveraging the CoT process itself as an avenue for oversight. By inspecting or supervising a model’s intermediate reasoning, we might detect misaligned

plans *in the act*. OpenAI observes that reading a model’s CoT process can reveal deceptive reasoning, and uses a separate oversight model - a CoT monitor - to flag such problematic reasoning [24]. However, a critical concern is that if models are directly penalized for showing deceptive thoughts, they might hide those thoughts rather than truly abandoning the deception.

In this work, we introduce CoT Monitor+, a framework that **integrates the SELF-MONITOR mechanism into the model’s reasoning process** to mitigate deceptive behavior. The core idea is to let the model generate a CoT while simultaneously evaluating it internally for signs of misalignment, i.e., *thinking before thinking*. Instead of depending solely on an external judge or separate oversight model, CoT Monitor+ empowers the model to actively participate in its own alignment oversight. Rather than relying solely on an external judge or a separate oversight model, CoT Monitor+ makes the model an active participant in its alignment oversight. By training the model to generate a specialized self-monitoring signal alongside its reasoning, we aim to achieve two goals: (1) **detect deceptive strategies as they emerge**, and (2) **discourage the model from pursuing such strategies before response**, as self-monitoring fosters an internal feedback loop that promotes honesty.

To our knowledge, this is the first framework to blend CoT reasoning with a built-in self-check for deceptive alignment. Overall, our contributions are summarized as follows:

- **Language models can self-monitor their reasoning processes** (Section 3). Building on this insight, we introduce SELF-MONITOR, a framework that enables models to oversee their own CoT reasoning within predefined safety protocols. The experimental results demonstrate that SELF-MONITOR substantially reduces deceptive alignment behaviors, achieving a 43.8% reduction compared to pure CoT reasoning.
- **Detect deceptive tendency in reasoning models with DECEPTIONBENCH** (Section 4). To mitigate the deceptive risks posed by advanced LLMs, we introduce DeceptionBench, the first systematic benchmark designed to assess deceptive behaviors behind reasoning models. DeceptionBench assesses model behavior across five categories, offering insights into when and how models may superficially appear aligned while internally pursuing misaligned goals.
- **SELF-MONITOR serve as reward signals in RL training** (Section 5). Relying solely on an external weak model’s CoT monitor as a reward signal for RL training can lead the model to strategically suppress its true intentions during the CoT process, resulting in more sophisticated forms of deception. In contrast, using the model’s own SELF-MONITOR feedback as the training signal encourages more faithful reasoning and reduces the incentive to conceal deceptive thoughts.

2 Preliminaries

We consider a large language model (LLM) parameterized by θ . The generation process of the LLM is defined by mapping an input context $\mathcal{X}$ to the output $\mathcal{Y}$ by $p(\cdot|\cdot; \theta)$. Unlike LLMs [25], reasoning models explicitly model the reasoning process. Rather than producing a one-shot output, reasoning models generate an intermediate sequence $\mathcal{Z}$ including both reasoning steps $\mathcal{R}$ and final answer $\mathcal{Y}$. Since the autoregressive language model generates tokens sequentially for a given position by relying solely on the previously generated token sequence, the model can be conceptualized as a Markov decision process (MDP) [26, 27], wherein the conditional probability $p(\mathcal{Z} | \mathcal{X}; \theta)$ is defined as,

$$p(\mathcal{Z} | \mathcal{X}; \theta) = p(\mathcal{R}, \mathcal{Y} | \mathcal{X}; \theta) = \prod_{t=1}^m p(z_t | z_{<t}, \mathcal{X}; \theta),$$

where $\mathcal{Z} = \mathcal{R} \oplus \mathcal{Y} = (z_1, \dots, z_m)$, z_i represents individual token from a predetermined vocabulary.

Supervised Fine-Tuning. SFT fine-tunes the pre-trained model using datasets sampled from a high-quality distribution, aligning the model with target tasks. For a SFT dataset $\mathcal{D}_{\text{SFT}} = \{(\mathcal{X}^{(i)}, \mathcal{Z}^{(i)})\}_{i=1}^N$, SFT aims to minimize the negative log-likelihood loss:

$$\mathcal{L}_{\text{SFT}}(\theta; \mathcal{D}_{\text{SFT}}) = -\mathbb{E}_{(\mathcal{X}, \mathcal{Z}) \sim \mathcal{D}_{\text{SFT}}} \log p(\mathcal{Z} | \mathcal{X}; \theta)$$

Reward Modeling. Rewards serve as the source of training signals in RL fine-tuning [27]. Preference-based [28] and rule-based [2] reward modeling are two representative approaches. In preference-based reward modeling, for a preference dataset $\mathcal{D}_R = \{(\mathcal{X}^{(i)}, \mathcal{Z}_w^{(i)}, \mathcal{Z}_l^{(i)})\}_{i=1}^N$ sampled from human preference distribution p^* , we estimate the reward model $r_\phi(\mathcal{Z}, \mathcal{X})$ parameters via maximum likelihood.